Chapter 3. Do-Calculus and Algorithmic Identification

Rules of do-calculus, completeness, c-components, the ID algorithm

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3.1 Do-calculus

Motivation. Do-calculus (Pearl 1995) gives three complete graphical rules for reducing $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ to an expression in $P(\mathbf{v})$ whenever this is possible.

3.1.1 Three examples via the truncated factorization

Motivation. Eliminate the $\mathbf{U}$-terms of a semi-Markovian model by hand.

Theorem 3.1 (Generalized truncated factorization)

In a semi-Markovian model $\mathcal{M}$, the distribution generated by $\mathrm{do}(\mathbf{X} = \mathbf{x})$ is

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u}),$$
$$\mathbf{V}' = \mathbf{V} \setminus \mathbf{X},$$

and for $\mathbf{Y} \subseteq \mathbf{V} \setminus \mathbf{X}$, $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{v} \setminus (\mathbf{x} \cup \mathbf{y})} \sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$.

Proof [Supplement]. Fix $\mathbf{u}$. In $\mathcal{M}_{\mathbf{x}}$ the mechanisms of $\mathbf{X}$ become constants and the others are unchanged, so $P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x}), \mathbf{u}) = \prod_{V_i \in \mathbf{V}'} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i)$, each factor the indicator of $f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i$ (acyclicity: solve in topological order). $P(\mathbf{u})$ is unchanged; average over $\mathbf{u}$. ■

Identification succeeds when the right-hand side can be rewritten without $\mathbf{U}$.

3.1.1.1 Back-door graph

$Z \to X$, $Z \to Y$, $X \to Y$, independent U_Z, U_X, U_Y. In $\mathcal{M}_x$: $Z \leftarrow f_Z(u_Z)$, $X \leftarrow x$, $Y \leftarrow f_Y(x, z, u_Y)$.

$$\begin{aligned} P(\mathbf{v}' \mid \mathrm{do}(x)) &= \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, z, u_Y) P(\mathbf{u}) \\ &= \Big(\sum_{u_Z} P(z \mid u_Z) P(u_Z) \Big) \Big(\sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y) \Big) \Big(\sum_{u_X} P(u_X) \Big) \\ &= P(z) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y \mid x, z) = P(z) P(y \mid x, z), \end{aligned}$$

using $U_Y \perp\!\!\!\perp \{X, Z\}$. Hence $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z) P(z)$.

3.1.1.2 Bow graph

$X \to Y$, $X \leftrightarrow Y$ through U. In $\mathcal{M}_x$: $X \leftarrow x$, $Y \leftarrow f_Y(x, u, u_Y)$.

$$P(y \mid \mathrm{do}(x)) = \sum_{u} P(u) \sum_{u_Y} P(y \mid x, u, u_Y) P(u_Y) \sum_{u_X} P(u_X) = \sum_{u} P(y \mid x, u) P(u).$$

$P(u)$ and $P(y \mid x, u)$ are not functions of $P(x, y)$, so U cannot be eliminated (observationally u is weighted by $P(u \mid x)$ instead).

3.1.1.3 Front-door graph

$X \to Z \to Y$, $X \leftrightarrow Y$ through U. In $\mathcal{M}_x$: $X \leftarrow x$, $Z \leftarrow f_Z(x, u_Z)$, $Y \leftarrow f_Y(z, u, u_Y)$.

$$P(z, y \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid x, u_Z) P(y \mid z, u, u_Y) P(\mathbf{u})$$

$$
\begin{aligned}
&= \Big(\sum_{u_Z} P(z \mid x, u_Z)P(u_Z)\Big)\Big(\sum_{u,u_Y} P(y \mid z, u, u_Y)P(u, u_Y)\Big) \\
&= P(z \mid x)\sum_{u} P(y \mid z, u)\sum_{x'} P(u \mid x')P(x') \\
&= P(z \mid x)\sum_{u,x'} P(y \mid z, u, x')P(u \mid x')P(x') && (Y \perp\!\!\!\perp X \mid Z, U) \\
&= P(z \mid x)\sum_{u,x'} P(y \mid z, u, x')P(u \mid x', z)P(x') && (U \perp\!\!\!\perp Z \mid X) \\
&= P(z \mid x)\sum_{x'}\sum_{u} P(y, u \mid z, x')P(x') \\
&= P(z \mid x)\sum_{x'} P(y \mid z, x')P(x').
\end{aligned}
$$

Therefore the **front-door formula**

$$P(y \mid \mathrm{do}(x)) = \sum_{z} P(z \mid x)\sum_{x'} P(y \mid z, x')P(x').$$

3.1.2 Identification approaches

Motivation. State the syntactic goal that do-calculus serves.

Goal: reduce $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ to terms estimable from $P(\mathbf{v})$, either via a $\mathbf{U}$-free truncated factorization (as above) or by rewriting Q into a *do-free* expression with equality-preserving rules, never mentioning $\mathbf{U}$.

3.1.3 Do-calculus

Motivation. Derive the three rules from three insights.

3.1.3.1 Prelude: independence in a submodel

$\mathrm{do}(\mathbf{X} = \mathbf{x})$ replaces each f_X by a constant: graphically $\mathcal{G}_{\overline{\mathbf{X}}}$, with $\mathbf{X}$ conditioned on. Hence

$$(\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}) \text{ under } \mathrm{do}(\mathbf{x}) \quad \Longleftarrow \quad (\mathbf{A} \perp_d \mathbf{B} \mid \mathbf{C}, \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}},$$

equivalently $(\mathbf{A} \perp_d \mathbf{B} \mid \mathbf{C})$ in $\mathcal{G} \setminus \mathbf{X}$ when $\mathbf{X}$ is disjoint from $\mathbf{A}, \mathbf{B}, \mathbf{C}$. E.g. in $A \to X \to D$ the intervention severs A from X and fixed X blocks every path through it.

3.1.3.2 Insight 1: adding/removing observations

Graph: $Z \to X \to Y$, $X \leftrightarrow Y$ (via U). Then

$$P(y, z \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid u_Z)P(y \mid x, u_Y, u)P(\mathbf{u}) = P(z)\sum_{u} P(y \mid x, u)P(u).$$

Marginalizing, $P(z \mid \mathrm{do}(x)) = P(z)\sum_u P(u)\sum_y P(y \mid x, u) = P(z)$, so

$$P(y \mid \mathrm{do}(x), z) = \frac{P(y, z \mid \mathrm{do}(x))}{P(z \mid \mathrm{do}(x))} = \sum_{u} P(y \mid x, u)P(u) = \sum_{z} P(y, z \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x)).$$

The reason is $(Z \perp_d Y \mid X)_{\mathcal{G}_{\overline{X}}}$. In $\mathcal{G}$, $Z \not\perp\!\!\!\perp Y$ (via $Z \to X \to Y$) and $Z \not\perp\!\!\!\perp Y \mid X$ (collider on $Z \to X \leftrightarrow Y$); in $\mathcal{M}_x$ the edge $Z \to X$ is cut, and $P(y \mid \mathrm{do}(x), z) = P_x(y \mid z)$ is an ordinary conditional there.

3.1.3.3 Insight 2: exchanging action and observation

Graph: $Z \to X$, $Z \to Y$, $X \to Y$. Given Z, Y is associated with X only through causal paths, so *seeing* $X = x$ equals *doing* $X = x$:

$$(Y \perp_d X \mid Z)_{\mathcal{G}_{\underline{X}}} \implies P(y \mid \mathrm{do}(x), z) = P(y \mid x, z).$$

Check: $P(y, z \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, z, u_Y) P(\mathbf{u}) = P(z) P(y \mid x, z)$ and $P(z \mid \mathrm{do}(x)) = \sum_y P(z) P(y \mid x, z) = P(z)$, so $P(y \mid \mathrm{do}(x), z) = P(z) P(y \mid x, z) / P(z) = P(y \mid x, z)$.

3.1.3.4 Insight 3: adding/removing actions

Graph: $X \to Y$, $Z \leftrightarrow Y$ (latent U_{zy}), $Z \leftrightarrow X$ (latent U_{zx}). With no causal path from X to Z,

$$(Z \perp_d X)_{\mathcal{G}_{\overline{X}}} \implies P(z \mid \mathrm{do}(x)) = P(z).$$

Check: $P(z \mid \mathrm{do}(x)) = \sum_y \sum_{u_{zy}, u_{zx}} P(z \mid u_{zy}, u_{zx}) P(y \mid x, u_{zy}) P(u_{zy}, u_{zx}) = \sum_{u_{zy}, u_{zx}} P(z \mid u_{zy}, u_{zx}) P(u_{zy}, u_{zx}) = P(z)$.

3.1.3.5 The rules

Notation: $\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$ is $\mathcal{G}$ with edges into $\mathbf{X}$ and out of $\mathbf{Z}$ removed.

Theorem 3.2 (Rules of do-calculus; Pearl 1995)

For any do-distribution induced by a causal model $\mathcal{M}$ with diagram $\mathcal{G}$, and disjoint $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{W}$:

Rule 1 (adding/removing observations) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

Rule 2 (action/observation exchange) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$.

Rule 3 (adding/removing actions) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}}$, where $\mathbf{Z}(\mathbf{W}) = \mathbf{Z} \setminus An(\mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

Proof [Supplement]. Work in $\mathcal{M}_{\mathbf{x}}$ with diagram $\mathcal{G}_{\overline{\mathbf{X}}}$; by the generalized truncated factorization its distribution (latents as explicit nodes) is Markov relative to $\mathcal{G}_{\overline{\mathbf{X}}}$, so d-separation there implies conditional independence in $P(\cdot \mid \mathrm{do}(\mathbf{x}))$.

Rule 1. $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$ gives $\mathbf{Y} \perp\!\!\!\perp \mathbf{Z} \mid \mathbf{W}$ in $P(\cdot \mid \mathrm{do}(\mathbf{x}))$ ($\mathbf{X}$ constant), which is the claim.

Augmented graph. Add to $\mathcal{G}_{\overline{\mathbf{X}}}$ a parent F_Z of each $Z \in \mathbf{Z}$: $F_Z = \text{idle}$ keeps the mechanism of Z, $F_Z = z'$ replaces it by z'. Then $P(\cdot \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}))$ is the law given $F_{\mathbf{Z}} = \mathbf{z}$ and $P(\cdot \mid \mathrm{do}(\mathbf{x}))$ the law given $F_{\mathbf{Z}} = \text{idle}$; the augmented model is Markov relative to $\mathcal{G}^F$.

Rule 2. On $\mathbf{Z} = \mathbf{z}$ the regimes $F_{\mathbf{Z}} = \mathbf{z}$ and $F_{\mathbf{Z}} = \text{idle}$ differ only in the mechanism of $\mathbf{Z}$, so it suffices that $\mathbf{Y} \perp\!\!\!\perp F_{\mathbf{Z}} \mid \mathbf{X}, \mathbf{Z}, \mathbf{W}$ in $\mathcal{G}^F$. A path from F_Z enters Z via $F_Z \to Z$; continuing out of Z, Z is a conditioned non-collider and the path is blocked; continuing into Z, Z is a conditioned collider. So the open paths are $F_Z \to Z \leftarrow \cdots$, open given $(\mathbf{X}, \mathbf{Z}, \mathbf{W})$ iff the corresponding path from Z to $\mathbf{Y}$ is open in $\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$. Hence $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$ implies $(F_{\mathbf{Z}} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{Z}, \mathbf{W})_{\mathcal{G}^F}$.

Rule 3. Now $\mathbf{Z}$ is unobserved and we need $\mathbf{Y} \perp\!\!\!\perp F_{\mathbf{Z}} \mid \mathbf{X}, \mathbf{W}$ in $\mathcal{G}^F$. A path from F_Z continues out of Z (unconditioned non-collider) or into Z (collider, open only if $Z \in An(\mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$). For $Z \in \mathbf{Z}(\mathbf{W})$ the second kind is blocked and the first is open given $(\mathbf{X}, \mathbf{W})$ exactly when the corresponding path is open in a graph where Z has no incoming edges, i.e. $\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}$. For $Z \in \mathbf{Z} \setminus \mathbf{Z}(\mathbf{W})$, separation in

$\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}$ blocks the paths through incoming edges too. So $(F_{\mathbf{Z}} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}^F}$ and both regimes give the same $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$. ■

With $\mathbf{W} = \emptyset$: Rule 1 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, Rule 2 $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$, Rule 3 $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{XZ}}}}$.

Simplified form. Let $\mathcal{H} = \mathcal{G} \setminus \mathbf{X}$ and $Q = P_{\mathbf{x}}$. Then

R1: $Q(\mathbf{y} \mid \mathbf{z}, \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}}$,

R2: $Q(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{z}, \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}_{\underline{\mathbf{Z}}}}$,

R3: $Q(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}_{\overline{\mathbf{Z(W)}}}}$, $\mathbf{Z(W)} = \mathbf{Z} \setminus An(\mathbf{W})_{\mathcal{H}}$.

Each rule concerns one fixed submodel $\mathcal{M}_{\mathbf{x}}$.

3.1.3.6 Understanding Rule 2

Since $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{x}, \mathbf{z}, \mathbf{w})$, Rule 2 asks whether model and submodel give the same conditional, e.g. $P(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathbf{z}, \mathbf{w})$.

Regime indicator. Augment $\mathcal{G}$ ($W \to Z$, $W \to Y$, $Z \to Y$) with $F_Z \to Z$:

$$f'_Z = \begin{cases} f_Z(W, U_Z) & \text{if } F_Z = 0, \\ z & \text{if } F_Z = 1. \end{cases}$$

Then $P(y \mid \mathrm{do}(z), z, w) = P(y \mid z, w)$ reads $P(y \mid z, w, F_Z = 1) = P(y \mid z, w, F_Z = 0)$, i.e. $(F_Z \perp\!\!\!\perp Y \mid Z, W)$ in $\mathcal{G}'$. With Z given, every d-connecting path is $F_Z \to Z \leftarrow \cdots Y$, so we check back-door paths of Z, i.e. Z versus Y in $\mathcal{G}_{\underline{Z}}$.

3.1.3.7 Understanding Rule 3

"Changing a function" subsumes "changing it to a constant", so

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) \iff P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), F_Z = 1, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), F_Z = 0, \mathbf{w}),$$

which holds if $F_{\mathbf{Z}} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}, \mathbf{X}$ in $\mathcal{G}'$. Paths $F_Z \to Z \to \cdots$ are always feasible; $F_Z \to Z \leftarrow \cdots$ only if $Z \in An(\mathbf{W})$ in $\mathcal{G}_{\overline{\mathbf{X}}}$. So edges into Z can be cut exactly for $Z \in \mathbf{Z(W)}$: condition $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}}$.

Example 3.1 (Why Z(W) is needed)

Graph: $T \to Z \to Y$, $Z \to W$. Is $P(t \mid \mathrm{do}(z), w) = P(t \mid w)$? In $\mathcal{G}_{\overline{Z}}$, T is isolated, so a naive check says yes. But $Z \in An(W)$, so $\mathbf{Z(W)} = \emptyset$ and in $\mathcal{G}$ itself $T \to Z$ remains: the condition fails. Indeed, in almost every compatible model,

$$P(t \mid \mathrm{do}(z), w) = \frac{P(t, w \mid \mathrm{do}(z))}{P(w \mid \mathrm{do}(z))} = \frac{\sum_y P(t)P(w \mid z)P(y \mid z)}{\sum_{y,t} P(t)P(w \mid z)P(y \mid z)}$$
$$= \frac{P(t)P(w \mid z)}{P(w \mid z)} = P(t) \neq P(t \mid w),$$

since observationally W informs on T through Z.

3.1.4 Do-calculus and identification

Motivation. Completeness, the front-door derivation, and tools for non-identifiability.

Theorem 3.3 (Completeness; Huang and Valtorta 2006, Shpitser and Pearl 2006)

$Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable from $P(\mathbf{v})$ and $\mathcal{G}$ iff there is a finite sequence of applications of the rules of do-calculus and the probability axioms that reduces Q to a do-free expression.

Proof [Supplement]. *If:* each rule and axiom preserves equality in every model compatible with $\mathcal{G}$, so a do-free reduction writes Q as a functional of $P(\mathbf{v})$. *Only if:* for identifiable Q the ID algorithm (Chapter 3, Section 3.2) returns an expression, each step being a do-calculus rule or probability axiom (Huang and Valtorta 2006; Shpitser and Pearl 2006); on failure it exhibits a hedge yielding two models with equal $P(\mathbf{v})$ and different Q (construction not reproduced). ■

Syntactic goal: re-express Q without do.

3.1.4.1 Front-door by do-calculus

Smoking $S \to$ Tar $T \to$ Cancer C, with $S \leftrightarrow C$.

$$
\begin{aligned}
P(c \mid \mathrm{do}(s)) &= \sum_t P(c \mid \mathrm{do}(s), t) P(t \mid \mathrm{do}(s)) && \text{probability axioms} \\
&= \sum_t P(c \mid \mathrm{do}(s), \mathrm{do}(t)) P(t \mid \mathrm{do}(s)) && \text{Rule 2: } (T \perp_d C \mid S)_{\mathcal{G}_{\overline{S}\underline{T}}} \\
&= \sum_t P(c \mid \mathrm{do}(t)) P(t \mid \mathrm{do}(s)) && \text{Rule 3: } (S \perp_d C \mid T)_{\mathcal{G}_{\overline{T}\,\overline{S}}} \\
&= \sum_t P(c \mid \mathrm{do}(t)) P(t \mid s) && \text{Rule 2: } (T \perp_d S)_{\mathcal{G}_{\underline{S}}} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', \mathrm{do}(t)) P(s' \mid \mathrm{do}(t)) && \text{probability axioms} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', t) P(s' \mid \mathrm{do}(t)) && \text{Rule 2: } (T \perp_d C \mid S)_{\mathcal{G}_{\underline{T}}} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', t) P(s') && \text{Rule 3: } (T \perp_d S)_{\mathcal{G}_{\overline{T}}}.
\end{aligned}
$$

Justifications: in $\mathcal{G}_{\overline{S}\underline{T}}$ only $S \leftrightarrow C$, $S \to T$ remain and S is given; in $\mathcal{G}_{\overline{T}\,\overline{S}}$, S connects only via $S \leftrightarrow C$, not opened by T; in $\mathcal{G}_{\underline{S}}$, $S \leftrightarrow C \leftarrow T$ is blocked at the collider C; in $\mathcal{G}_{\underline{T}}$, $T \leftarrow S \leftrightarrow C$ is blocked by S; in $\mathcal{G}_{\overline{T}}$, T reaches S only via the collider C.

3.1.4.2 Proving non-identifiability

Example 3.2

Bow graph $X \to Y$, $X \leftrightarrow Y$, U a fair coin, $\mathcal{F}$: $X \leftarrow U$, $Y \leftarrow f_Y(X, U)$. Two models agree with $\mathcal{F}$ when $U = X$ (the only observed case) and differ off the diagonal:

$$
\mathcal{F}_x^{(1)}: \; Y \leftarrow \begin{cases} f_Y(X,U) & U = X \\ 0 & \text{otherwise} \end{cases},
$$

$$
\mathcal{F}_x^{(2)}: \; Y \leftarrow \begin{cases} f_Y(X,U) & U = X \\ 1 & \text{otherwise.} \end{cases}
$$

Under $\mathrm{do}(X = x)$, U and X need not match:

$$
P^{(i)}(Y = 1 \mid \mathrm{do}(x)) = \sum_u P^{(i)}(y \mid x, u) P(u) = P(y \mid x) P(x) + \mathbb{1}_{i=2}\,(1 - P(x)).
$$

Same $\mathcal{G}$ and $P(\mathbf{v})$, different effects whenever $P(x) < 1$.

Lemma 3.4 (Graph–subgraph; Tian and Pearl 2002)

If $Q = P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is not identifiable in $\mathcal{G}$, it is not identifiable in any graph obtained by adding a directed or bidirected edge to $\mathcal{G}$. Conversely, if Q is identifiable in $\mathcal{G}$, it remains identifiable after removing edges.

Proof [Supplement]. Every model compatible with $\mathcal{G}$ is compatible with $\mathcal{G}^+ = \mathcal{G}$ plus an edge (functions may ignore a new parent; a new latent may have no effect). So two $\mathcal{G}$-models with equal $P(\mathbf{v})$ and different Q also witness non-identifiability in $\mathcal{G}^+$. The converse is the contrapositive for $\mathcal{G}^- \subseteq \mathcal{G}$. ■

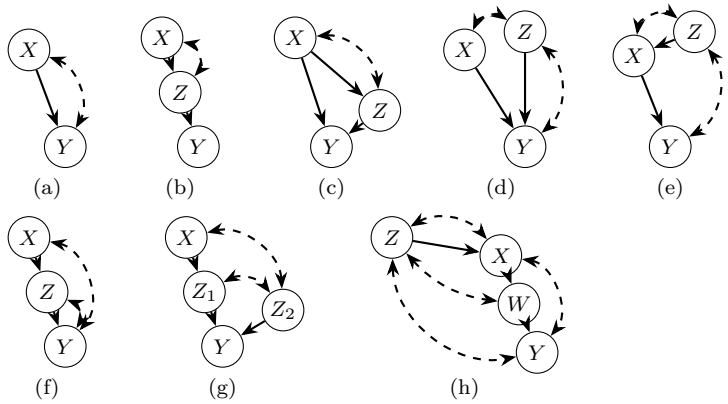
Theorem 3.5 (Non-identifiability criterion, singleton; Tian and Pearl 2002)

If there is a bidirected path connecting X to any of its children in $\mathcal{G}$, then $P(\mathbf{v}' \mid \text{do}(x))$ is not identifiable from $P(\mathbf{v})$ and $\mathcal{G}$.

Proof [Supplement]. *Base case.* $X \to Y$, $X \leftrightarrow Y$, binary U, X, Y. Model 1: $X = U$, $Y = X$; model 2: $X = U$, $Y = U$. Both give $P(x, y) = P(X = Y = x) = P(U = x)$, but $P(Y = 1 \mid \text{do}(X = 1))$ is 1 versus $P(U = 1)$. *General case.* For $X \leftrightarrow W_1 \leftrightarrow \cdots \leftrightarrow W_k \leftrightarrow Y$ with Y a child, let every latent on the path equal a common binary U and every W_j copy U, other variables independent noise. The two models still agree on $P(\mathbf{v})$ and disagree on $P(\mathbf{v}' \mid \text{do}(x))$; by the graph–subgraph lemma the remaining edges of $\mathcal{G}$ preserve this (Tian and Pearl 2002). ■

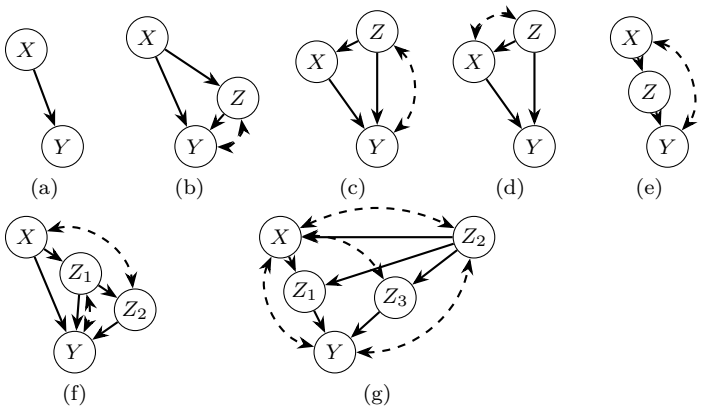
3.1.4.3 Catalogue of graphs

Graphs in which $P(y \mid \text{do}(x))$ is not identifiable (Pearl 2000, Fig. 3.9):



(a) is the bow graph, while (b) and (c) contain the bow $X \to Z$, $X \leftrightarrow Z$; in (d)–(h) X is joined to a child by a bidirected path (e.g. (d) $X \leftrightarrow Z \leftrightarrow Y$; (g) $X \leftrightarrow Z_2 \leftrightarrow Z_1$, Z_1 a child of X).

Graphs in which $P(y \mid \text{do}(x))$ is identifiable (Pearl 2000, Fig. 3.8):



E.g. in (b) X has no incoming edges, so $P(y \mid \mathrm{do}(x)) = P(y \mid x)$;

(c) and (d) follow by back-door adjustment for Z, and (e) by the front-door formula.

3.1.5 Back to Simpson's paradox

Motivation. The do-operator dissolves the paradox, which statistics alone cannot.

3.1.5.1 Pearson's shock

Pearson, Lee and Bramley-Moore (1899): mixing heterogeneous groups without "organic" correlation can produce a *spurious* correlation, so correlations always carry an error of unknown size. The same data give opposite conclusions aggregated or stratified; resolving this requires causal reasoning.

Examples. (1) Berkeley admissions (Bickel et al., *Science* 1975): overall 44% of men vs. 30% of women admitted, yet within departments a slight bias favoring women, who applied to more competitive departments. Whether to condition on department depends on the causal model (e.g. latent confounding of department and admission).

(2) "Reverse regression" in salary-discrimination cases: compare salaries of equally qualified men and women, or qualifications of equally paid ones? (Pearl 2009, §4.5; Bhadane et al. 2025.)

3.1.5.2 Dissolving the paradox with the do-operator

The tables of Chapter 1 give

$$P(Y \mid F, X) < P(Y \mid F, \neg X), \qquad P(Y \mid \neg F, X) < P(Y \mid \neg F, \neg X), \qquad P(Y \mid X) > P(Y \mid \neg X).$$

If instead

$$P(Y \mid F, \mathrm{do}(X)) < P(Y \mid F, \mathrm{do}(\neg X)), \qquad P(Y \mid \neg F, \mathrm{do}(X)) < P(Y \mid \neg F, \mathrm{do}(\neg X)),$$

and the action does not affect F (sex), then $P(Y \mid \mathrm{do}(X)) < P(Y \mid \mathrm{do}(\neg X))$, since by Rule 3 ($F$ not a descendant of X)

$$P(y \mid \mathrm{do}(x)) = \sum_f P(y \mid \mathrm{do}(x), f) P(f \mid \mathrm{do}(x)) = \sum_f P(y \mid \mathrm{do}(x), f) P(f),$$

and both sides average the stratum effects with the *same* weights $P(f)$.

3.1.5.3 The paradox stems from causal interpretation

Surprise arises only about efficacy, not prediction: when two causal models fit the same data but recommend differently, no statistical criterion picks the right table.

Theorem 3.6 (Sure-thing principle; Savage 1972, Pearl 2000 Thm. 6.1.1)

An action X that increases the probability of an event Y in each subpopulation must also increase the probability of Y in the population as a whole, provided that the action does not change the distribution of the subpopulations.

Proof [Supplement]. Let F index the subpopulations, $P(f \mid \mathrm{do}(x)) = P(f \mid \mathrm{do}(\neg x)) = P(f)$. Then

$$P(Y \mid \mathrm{do}(x)) = \sum_f P(Y \mid \mathrm{do}(x), f)P(f) > \sum_f P(Y \mid \mathrm{do}(\neg x), f)P(f) = P(Y \mid \mathrm{do}(\neg x)),$$

term by term with identical weights, so no reversal is possible. ■

Reversal is possible in the calculus of proportions, impossible in the calculus of causation. Applications: adjustment, salary discrimination, optimal treatment design, fairness in ML.

3.1.5.4 Appendix: the Berkeley data

Department	Men applied	Men admitted	Women applied	Women admitted
A	825	62%	108	82%
B	560	56%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%
Total	2691	44%	1835	30%

Women applied disproportionately to competitive departments (C–F). Which graph?

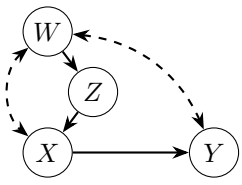
- $\mathcal{G}_1$: Sex $\to$ Dept $\to$ Admit, Sex $\to$ Admit (Dept a mediator). The total effect is the aggregate rate; adjusting for Dept gives the controlled direct effect (discrimination within departments), the indirect effect being sex's influence on department choice.
- $\mathcal{G}_2$: $U \to$ Dept, $U \to$ Sex, Dept $\to$ Admit, Sex $\to$ Admit, U latent (Dept a confounder). Dept blocks Sex $\leftarrow U \to$ Dept $\to$ Admit, so $P(\text{admit} \mid \mathrm{do}(\text{sex})) = \sum_d P(\text{admit} \mid \text{sex}, d)P(d)$, showing no significant discrimination.

Same data and adjustment, different causal interpretations.

3.1.6 Summary and food for thought

- Do-calculus characterizes policy evaluation for atomic interventions syntactically.
- Non-parametric confounding and identification are essentially solved.
- Simpson's paradox is mathematised and dissolved.
- Applications: social and health sciences, statistics, ML, AI.

Food for thought: the napkin graph. $W \to Z \to X \to Y$, $W \leftrightarrow X$, $W \leftrightarrow Y$. Is $P(y \mid \mathrm{do}(x))$ identifiable?



Yes, though neither back-door nor front-door applies: intervening on Z makes X unconfounded given W,

$$P(y \mid \mathrm{do}(x)) = \frac{\sum_w P(x, y \mid z, w)P(w)}{\sum_w P(x \mid z, w)P(w)} \qquad \text{(for any } z\text{)}.$$

(Rule 3: $P(y \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x), \mathrm{do}(z))$; back-door through W: $P(x, y \mid \mathrm{do}(z)) = \sum_w P(x, y \mid z, w)P(w)$; Rule 2: $P(y \mid \mathrm{do}(z), \mathrm{do}(x)) = P(y \mid \mathrm{do}(z), x)$, since in $\mathcal{G}_{\overline{Z}\underline{X}}$ every X–Y path goes through W, cut from X except via Z. Section 3.2 does this systematically.)

3.2 Algorithmic identification

Motivation. Decide identifiability of $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ mechanically, without searching for do-calculus steps.

Roadmap. (1) Factorize the distribution along the causal diagram (c-components, c-factors).

(2) Operations that identify particular factors.

(3) Write the target in factors and identify each one.

3.2.1 Factorizing observational distributions

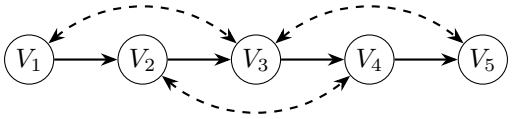
Motivation. Group variables sharing latent confounders; the joint factorizes over these groups.

3.2.1.1 Confounded components

Definition (C-component)

When V_i and V_j share an unobservable parent U, they are in the same *confounded component* (c-component). Closing under chaining: two variables are in the same c-component iff they are connected by a path of bidirected edges.

Running example: $V_1 \to V_2 \to V_3 \to V_4 \to V_5$ with U_1: $V_1 \leftrightarrow V_3$, U_2: $V_3 \leftrightarrow V_5$, U_3: $V_2 \leftrightarrow V_4$.



The relation is an equivalence (read it off the bidirected edges alone), so it *partitions* the observed variables: here $\{V_1, V_3, V_5\}$ and $\{V_2, V_4\}$.

3.2.1.2 Markovian case

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

For $Z \to X$, $Z \to Y$, $X \to Y$:

$$\begin{aligned} P(z, x, y) &= \sum_{u_z} P(z \mid u_z)P(u_z) \sum_{u_x} P(x \mid z, u_x)P(u_x) \sum_{u_y} P(y \mid x, z, u_y)P(u_y) \\ &= P(z)P(x \mid z)P(y \mid x, z). \end{aligned}$$

Each canonical factor $P(v_i \mid \mathbf{pa}_i)$ is computable from $P(\mathbf{v})$:

$$P(v_i \mid \mathbf{pa}_i) = \frac{\sum_{\mathbf{v}\setminus(\{v_i\}\cup\mathbf{pa}_i)} P(\mathbf{v})}{\sum_{\mathbf{v}\setminus\mathbf{pa}_i} P(\mathbf{v})},$$

$$\text{e.g. } P(z) = \sum_{x,y} P(\mathbf{v}),\ P(x \mid z) = \frac{\sum_y P(\mathbf{v})}{\sum_{x,y} P(\mathbf{v})},\ P(y \mid x, z) = \frac{P(\mathbf{v})}{\sum_y P(\mathbf{v})}.$$

3.2.1.3 Semi-Markovian case

With only U_1 ($V_1 \leftrightarrow V_3$) and U_2 ($V_3 \leftrightarrow V_5$):

$$P(\mathbf{v}) = P(v_2 \mid v_1)P(v_4 \mid v_3) \sum_{u_1,u_2} P(u_1,u_2)P(v_1 \mid u_1)P(v_3 \mid v_2,u_1,u_2)P(v_5 \mid v_4,u_2).$$

Adding U_3 ($V_2 \leftrightarrow V_4$):

$$\begin{aligned} P(\mathbf{v}) = &\Big[\sum_{u_3} P(v_2 \mid v_1,u_3)P(v_4 \mid v_3,u_3)P(u_3)\Big] \\ &\cdot \Big[\sum_{u_1,u_2} P(u_1,u_2)P(v_1 \mid u_1)P(v_3 \mid v_2,u_1,u_2)P(v_5 \mid v_4,u_2)\Big]. \end{aligned}$$

The sums separate because different c-components share no latent variable.

3.2.1.4 C-factors

For $C \subseteq \mathbf{V}$ define

$$Q[C](\mathbf{c},\mathbf{pa}_C) = \sum_{\mathbf{u}(C)} P(\mathbf{u}(C)) \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i),$$

$$\mathbf{U}(C) = \bigcup_{V_i \in C} \mathbf{U}_i.$$

Then

$$P(\mathbf{v}) = Q[V_2,V_4](v_2,v_4,v_1,v_3) \cdot Q[V_1,V_3,V_5](v_1,v_3,v_5,v_2,v_4),$$

abbreviated $P(\mathbf{v}) = Q[V_1,V_3,V_5]\,Q[V_2,V_4]$, with no $\mathbf{U}$. $Q[\mathbf{V}] = P(\mathbf{v})$ and $Q[\emptyset] = 1$.

3.2.1.5 C-factors are causal effects

For $C \subseteq \mathbf{V}$, by the truncated factorization,

$$P(\mathbf{c} \mid \mathrm{do}(\mathbf{v}\setminus\mathbf{c})) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) = \sum_{\mathbf{u}(C)} \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u}(C)) = Q[C],$$

since latents that are not parents of C sum out. *C-factors are interventional distributions.* E.g.

$$Q[V_1,V_3,V_5](v_1,v_3,v_5,v_2,v_4) = \sum_{u_1,u_2} P(u_1,u_2)P(v_1 \mid u_1)P(v_3 \mid v_2,u_1,u_2)P(v_5 \mid v_4,u_2)$$

is the law of (V_1,V_3,V_5) under $\mathrm{do}(V_2 = v_2, V_4 = v_4)$.

3.2.2 C-factor algebra

Motivation. Two operations drive identification: marginalizing to ancestral sets and factorizing into c-components.

3.2.2.1 Marginalizing variables in c-factors

In $Q[V_1, V_3, V_5]$ no member is a parent of another, so each appears in one term and

$$\sum_{v_3} Q[V_1, V_3, V_5] = \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) \Big(\sum_{v_3} P(v_3 \mid v_2, u_1, u_2) \Big) P(v_5 \mid v_4, u_2) = Q[V_1, V_5].$$

In $Q[V_1, V_2, V_3]$, V_1 is a parent of V_2:

$$\sum_{v_1} Q[V_1, V_2, V_3] = \sum_{\mathbf{u}} P(\mathbf{u}) P(v_3 \mid v_2, u_1, u_2) \sum_{v_1} P(v_1 \mid u_1) P(v_2 \mid v_1, u_3),$$

whereas

$$Q[V_2, V_3](v_2, v_3, v_1) = \sum_{\mathbf{u}} P(\mathbf{u}) P(v_3 \mid v_2, u_1, u_2) P(v_2 \mid v_1, u_3)$$

keeps v_1 fixed. The coupling of v_1 with u_1 (shared with V_3) gives $\sum_{v_1} Q[V_1, V_2, V_3] \neq Q[V_2, V_3]$ in general: only variables not feeding the rest can be summed out.

Lemma 3.7 (Ancestral reduction)

Let $W \subseteq C \subseteq \mathbf{V}$. If W is ancestral in $\mathcal{G}[C]$ (it contains all its ancestors within the subgraph induced by C), then

$$Q[W] = \sum_{\mathbf{c} \setminus \mathbf{w}} Q[C].$$

Proof [Supplement]. Sum out $C \setminus W$ in reverse topological order. When $V \in C \setminus W$ is summed it has no remaining child: a child in W would make $V \in W$ (W ancestral), and children in $C \setminus W$ are already summed. So v appears in $Q[C] = \sum_{\mathbf{u}} \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$ only in its own factor, which sums to one; latents attached only to summed variables integrate out. What remains is $Q[W]$. ■

Example: in $\mathcal{G}[C]$, $C = \{V_1, V_2, V_3\}$ ($V_1 \to V_2 \to V_3$), $\{V_1, V_2\}$ and $\{V_1\}$ are ancestral, $\{V_2, V_3\}$ is not.

3.2.2.2 C-component factorization

Lemma 3.8 (Tian 2002)

For any $H \subseteq \mathbf{V}$, let $H_1, \ldots, H_k$ be the c-components of $\mathcal{G}[H]$. Then

$$Q[H] = \prod_{j=1}^{k} Q[H_j].$$

Moreover, let $V^{(1)} < \cdots < V^{(|H|)}$ be a topological order of $\mathcal{G}[H]$ and $H^{\leq i}$ the variables of H up to and including $V^{(i)}$. Then each factor is computable from $Q[H]$:

$$Q[H_j] = \prod_{V^{(i)} \in H_j} \frac{Q[H^{\leq i}]}{Q[H^{\leq i-1}]}, \qquad Q[H^{\leq i}] = \sum_{\mathbf{h} \setminus \mathbf{h}^{\leq i}} Q[H].$$

Proof [Supplement]. A latent with children in two c-components would join them, so the latents split by c-component. In $Q[H] = \sum_{\mathbf{u}} \prod_{V_i \in H} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$ the latents of H_j appear only in factors of H_j and the groups are independent under $P(\mathbf{u})$, so the sum factorizes into $\prod_j Q[H_j]$. For the second part, $H^{\leq i}$ is ancestral in $\mathcal{G}[H]$, so $Q[H^{\leq i}] = \sum_{h \setminus h^{\leq i}} Q[H]$ by ancestral reduction; the

ratios $Q[H^{\le i}]/Q[H^{\le i-1}]$ are functions of $Q[H]$, and their product over H_j reproduces $Q[H_j]$ (each latent group integrated once). ■

Example 3.3 (Factorizing $P(\mathbf{v})$ by c-components)

Graph: $Z_2 \to X$, $Z_2 \to Z_1$, $Z_2 \to Z_3$, $X \to Z_1$, $Z_1 \to Y$, $Z_3 \to Y$, $Z_2 \leftrightarrow X$, $X \leftrightarrow Z_3$, $X \leftrightarrow Y$, $Z_2 \leftrightarrow Y$. C-components $\{Z_2, Z_3, X, Y\}$ and $\{Z_1\}$:

$$Q[\mathbf{V}] = Q[Z_2, Z_3, X, Y]\, Q[Z_1],$$
$$P(\mathbf{v}) = P(z_2, z_3, x, y \mid \mathrm{do}(z_1))\, P(z_1 \mid \mathrm{do}(z_2, z_3, x, y)) = P(z_2, z_3, x, y \mid \mathrm{do}(z_1))\, P(z_1 \mid \mathrm{do}(z_2, x)),$$

the last step by Rule 3 (only the parents Z_2, X of Z_1 matter).

Example 3.4 (Running example)

With $H = \mathbf{V}$ and the order $V_1 < \cdots < V_5$, $Q[\mathbf{V}] = Q[V_1, V_3, V_5]Q[V_2, V_4]$ and

$$\begin{aligned} Q[V_1, V_3, V_5] &= \frac{Q[V_1]}{Q[\emptyset]} \cdot \frac{Q[V_1, V_2, V_3]}{Q[V_1, V_2]} \cdot \frac{Q[V_1, \ldots, V_5]}{Q[V_1, \ldots, V_4]} \\ &= \frac{\sum_{v_2,\ldots,v_5} P(\mathbf{v})}{\sum_{\mathbf{v}} P(\mathbf{v})} \cdot \frac{\sum_{v_4,v_5} P(\mathbf{v})}{\sum_{v_3,v_4,v_5} P(\mathbf{v})} \cdot \frac{P(\mathbf{v})}{\sum_{v_5} P(\mathbf{v})} \\ &= \frac{P(v_1)}{1} \cdot \frac{P(v_1, v_2, v_3)}{P(v_1, v_2)} \cdot \frac{P(\mathbf{v})}{P(v_1, \ldots, v_4)} = P(v_1)P(v_3 \mid v_1, v_2)P(v_5 \mid v_1, \ldots, v_4), \\ Q[V_2, V_4] &= \frac{Q[V_1, V_2]}{Q[V_1]} \cdot \frac{Q[V_1, \ldots, V_4]}{Q[V_1, V_2, V_3]} = P(v_2 \mid v_1)P(v_4 \mid v_1, v_2, v_3). \end{aligned}$$

$\mathcal{G}[\{V_2, V_4\}]$ has no directed edges, so $\{V_2\}$ and $\{V_4\}$ are ancestral there:

$$Q[V_2] = \sum_{v_4} Q[V_2, V_4] = P(v_2 \mid v_1) \sum_{v_4} P(v_4 \mid v_1, v_2, v_3) = P(v_2 \mid v_1),$$
$$Q[V_4] = \sum_{v_2} P(v_2 \mid v_1)P(v_4 \mid v_1, v_2, v_3).$$

3.2.2.3 Visualization

From $Q[\mathbf{V}]$ one moves down a lattice of c-factors by summing out a "last" variable ($Q[\mathbf{V}] \to Q[V_{1234}] \to Q[V_{123}] \to Q[V_{12}] \to Q[V_1]$) or splitting along c-components ($Q[\mathbf{V}] \to Q[V_{135}] \cdot Q[V_{24}]$, $Q[V_{1234}] \to Q[V_{13}] \cdot Q[V_{24}]$, $Q[V_{123}] \to Q[V_{13}] \cdot Q[V_2]$). From $Q[V_{135}]$ every subset is reachable by summation (no directed edges in $\mathcal{G}[\{V_1, V_3, V_5\}]$), and $Q[V_{15}] = Q[V_1]Q[V_5]$ since V_1, V_5 are not bidirected-connected within $\{V_1, V_5\}$.

Summary of the algebra.

1. *Reduce to an ancestral set:* $Q[W] = \sum_{\mathbf{c}\setminus\mathbf{w}} Q[C]$ if W is ancestral in $\mathcal{G}[C]$.
2. *Factorize into c-components:* $Q[H] = \prod_j Q[H_j]$, with $H_1, \ldots, H_k$ the c-components of $\mathcal{G}[H]$, each computable from $Q[H]$.

Together they yield an if-and-only-if condition for identifiability.

3.2.3 Causal queries in terms of c-factors: examples

Motivation. Rederive the back-door, front-door and napkin formulas with c-factor algebra.

3.2.3.1 Back-door

$Z \to X$, $Z \to Y$, $X \to Y$ (Markovian): $P(\mathbf{v}) = Q[X]Q[Y]Q[Z]$. In $\mathcal{G}_{\overline{X}}$, $P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$ with

$$Q[Y] = \frac{Q[X,Y,Z]}{Q[X,Z]} = \frac{P(\mathbf{v})}{\sum_y P(\mathbf{v})} = P(y \mid x, z), \qquad Q[Z] = \sum_{x,y} Q[X,Y,Z] = P(z),$$

so $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid z, x)P(z)$.

3.2.3.2 Front-door

$X \to Z \to Y$, $X \leftrightarrow Y$: $P(\mathbf{v}) = Q[X,Y] \cdot Q[Z]$ with (order $X < Z < Y$)

$$Q[X,Y] = \frac{Q[X]}{Q[\emptyset]} \cdot \frac{Q[X,Z,Y]}{Q[X,Z]} = P(x)P(y \mid x, z), \qquad Q[Z] = \frac{Q[X,Z]}{Q[X]} = P(z \mid x).$$

Then

$$P(y \mid \mathrm{do}(x)) = \sum_z P(y, z \mid \mathrm{do}(x)) = \sum_z Q[Y,Z] = \sum_z Q[Y]Q[Z],$$

and $\{Y\}$ is ancestral in $\mathcal{G}[\{X,Y\}]$, so $Q[Y] = \sum_{x'} Q[X,Y] = \sum_{x'} P(x')P(y \mid x', z)$. Hence $P(y \mid \mathrm{do}(x)) = \sum_z P(z \mid x) \sum_{x'} P(x')P(y \mid x', z)$.

3.2.3.3 Napkin

$W_1 \to W_2 \to X \to Y$, $W_1 \leftrightarrow X$, $W_1 \leftrightarrow Y$. Then $P(\mathbf{v}) = Q[W_1, X, Y]\, Q[W_2]$ and

$$P(y \mid \mathrm{do}(x)) = \sum_{w_1, w_2} Q[W_1, W_2, Y] = Q[Y],$$

as $\{Y\}$ is ancestral in $\mathcal{G}[\{W_1, W_2, Y\}]$. With the order $W_1 < W_2 < X < Y$,

$$Q[W_1, X, Y] = \frac{Q[W_1]}{Q[\emptyset]} \cdot \frac{Q[W_1, W_2, X]}{Q[W_1, W_2]} \cdot \frac{Q[W_1, W_2, X, Y]}{Q[W_1, W_2, X]} = P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x).$$

$\{X,Y\}$ is ancestral in $\mathcal{G}[\{W_1, X, Y\}]$, so $Q[X,Y] = \sum_{w_1} P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x)$; in $\mathcal{G}[\{X,Y\}]$, X and Y are not bidirected-connected, so $Q[X,Y] = Q[X]Q[Y]$ and

$$P(y \mid \mathrm{do}(x)) = Q[Y] = \frac{Q[X,Y]}{Q[X]} = \frac{Q[X,Y]}{\sum_y Q[X,Y]} = \frac{\sum_{w_1} P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x)}{\sum_{w_1} P(w_1)P(x \mid w_1, w_2)}.$$

The same result by do-calculus.

$$\begin{aligned}
P(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x, w_1, w_2)) && \text{Rule 3} \\
&= P(y \mid x, \mathrm{do}(w_1, w_2)) && \text{Rule 2} \\
&= \frac{P(y, x \mid \mathrm{do}(w_1, w_2))}{P(x \mid \mathrm{do}(w_1, w_2))} = \frac{P(y, x \mid \mathrm{do}(w_2))}{P(x \mid \mathrm{do}(w_2))} && \text{Rule 3} \\
&= \frac{\sum_{w_1} P(y, x \mid \mathrm{do}(w_2), w_1)P(w_1 \mid \mathrm{do}(w_2))}{\sum_{w_1} P(x \mid \mathrm{do}(w_2), w_1)P(w_1 \mid \mathrm{do}(w_2))} \\
&= \frac{\sum_{w_1} P(y, x \mid w_2, w_1)P(w_1)}{\sum_{w_1} P(x \mid w_2, w_1)P(w_1)} && \text{Rules 2 and 3.}
\end{aligned}$$

3.2.3.4 Overview of the three derivations

Back-door	Front-door	Napkin
$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[X]\,Q[Y]\,Q[Z]$	$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[X,Y]\,Q[Z]$	$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[W_1,X,Y]\,Q[W_2]$
	$Q[X,Y] \xrightarrow{\text{sum}} Q[Y]$	$Q[W_1,X,Y] \xrightarrow{\text{sum}} Q[X,Y] \xrightarrow{\text{fact.}} Q[X]\,Q[Y]$
$P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$	$P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$	$P(y \mid \mathrm{do}(x)) = Q[Y]$

3.2.4 A general identification algorithm

Motivation. Combine the two operations into a sound and complete procedure.

3.2.4.1 Reducing the query to c-factors

By Rule 3, every variable that is not an ancestor of $\mathbf{Y}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$ can be intervened on without changing the query. Let $C = An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}}$. Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{c}\setminus\mathbf{y}} P(\mathbf{c} \mid \mathrm{do}(\mathbf{v}\setminus\mathbf{c})) = \sum_{\mathbf{v}\setminus(\mathbf{x}\cup\mathbf{y})} Q[\mathbf{V}\setminus\mathbf{X}] = \sum_{\mathbf{c}\setminus\mathbf{y}} Q[C].$$

If $\mathcal{G}[C]$ has c-components $C_1,\dots,C_k$, then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{c}\setminus\mathbf{y}} \prod_{i=1}^{k} Q[C_i],$$

while $P(\mathbf{v}) = \prod_j Q[T_j]$ over the c-components T_j of $\mathcal{G}$, each $Q[T_j]$ computable. Each C_i lies in a single T_j; it remains to decide whether $Q[C_i]$ is computable from $Q[T_j]$.

3.2.4.2 The Identify procedure

Algorithm 1 IDENTIFY$(C, T, Q[T], \mathcal{G})$ (decides whether $Q[C]$ is computable from $Q[T]$, $C \subseteq T$)

```
A ← An(C)_{G[T]}
if A = C then                                   ▷ C is ancestral in T
    return Q[C] = Σ_{t\c} Q[T]                  ▷ marginalize
else if A = T then                              ▷ nothing to marginalize, unable to factorize
    return FAIL
else
    A_i ← the c-component of G[A] containing C
    obtain Q[A] = Σ_{t\a} Q[T], then Q[A_i] by c-component factorization   ▷ marginalize, then factorize
    return IDENTIFY(C, A_i, Q[A_i], G)
```

The three cases. (1) $A = C$: marginalize, $Q[C] = \sum_{\mathbf{t}\setminus\mathbf{c}} Q[T]$.

(2) $A = T$: T is its own ancestral closure and a single c-component, so it cannot shrink: fail.

(3) $C \subsetneq A \subsetneq T$: marginalize to A, factorize, keep the c-component $A_i \supseteq C$ (C is a c-component of $\mathcal{G}[C]$), and recurse. $T \supsetneq A_i \supsetneq \cdots$, so at most $|T|$ steps.

Theorem 3.9 (Completeness; Tian 2002)

$P(\mathbf{y} \mid \text{do}(\mathbf{x}))$ is identifiable from $\mathcal{G}$ and $P(\mathbf{v})$ if and only if each c-factor $Q[C_i]$ is identified by $\textsc{Identify}(C_i, T_i, Q[T_i], \mathcal{G})$, where T_i is the c-component of $\mathcal{G}$ containing C_i.

Proof [Supplement]. *If:* each call returns an expression in $Q[T_i]$, identified from $P(\mathbf{v})$ by c-component factorization ($H = \mathbf{V}$); combining the factors by the generalized truncated factorization gives $P(\mathbf{y} \mid \text{do}(\mathbf{x}))$. *Only if:* a failure $A = T$, $C_i \subsetneq T$ contains a hedge, from which Shpitser and Pearl (2006) build two models agreeing on $P(\mathbf{v})$ but not on the query (construction not reproduced). ■

Huang and Valtorta (2006) and Shpitser and Pearl (2006) proved completeness of the equivalent ID algorithm and, through it, of do-calculus.

Example 3.5 (Bow graph fails)

$X \to Y$, $X \leftrightarrow Y$, query $P(y \mid \text{do}(x))$: $C = An(Y)_{\mathcal{G}_{\overline{X}}} = \{Y\}$, $T = \{X, Y\}$, and $A = An(Y)_{\mathcal{G}[T]} = \{X, Y\} = T$: fail, matching Chapter 2.

Example 3.6 (Front-door succeeds)

$X \to Z \to Y$, $X \leftrightarrow Y$: $C = \{Z, Y\}$ with c-components $\{Z\}$, $\{Y\}$. $Q[Z]$ is a c-component of $\mathcal{G}$. For $Q[Y]$, $T = \{X, Y\}$ and $A = An(Y)_{\mathcal{G}[T]} = \{Y\} = C$ ($Z \notin T$), so $Q[Y] = \sum_x Q[X, Y]$.